

BY 528 : Sensory Biology

Course Code : BY 528

Course Name : Sensory Biology

L-T-P-C : 3-0-0-3

Intended for : Elective for B.Tech, M.Tech and Ph.D.

Prerequisite : IC136 - Understanding Biotechnology & its Applications or Consent of faculty member

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Introduction to the Sensory System:** Overview of sensory systems in living organisms. Types of sensory stimuli (e.g., light, sound, touch, taste, smell) in humans and non-human organisms (eg. plants, insects, fishes, bacteria, fungi etc.). (xx hrs)
- **Anatomy of the Sensory System:** Human brain anatomy with focus on sensory lobes. Structure and function of the major sensory organs (e.g. eyes, ears, skin, tongue, nose). Cells and tissues involved in sensory system and their properties. Neural pathways for transmitting sensory information to the brain. (xx hrs)
- **Neuron and Action potential:** Basic cell biology of neuron. The process of converting sensory stimuli into electrical signals. Sensory receptor cells and their properties. Molecular mechanisms of sensory transduction. Sensory adaptation and plasticity. (xx hrs)
- **Functional Significance of the Sensory System:** Evolutionary origins and adaptive significance of sensory systems, sensory ecology and behavior, sensory system disorders and diseases. (xx hrs)
- **Research and Applications of Sensory Biology:** Tools for studying and modulation of sensory systems. (xx hrs)
- **Electronic equivalent of human senses:** Bioinspired electronics for artificial sensory systems. (xx hrs)

Textbooks:

1. Liqun Luo, **Principles of Neurobiology**, Garland Science. <https://web.math.princeton.edu/~sswang/Principles-of-Neurobiology.pdf>

References:

1. Eric Kandel, James Schwartz, and Thomas Jessell, **Principles of Neural Science**.
2. <https://openstax.org/books/biology/pages/36-introduction>
3. Related journal articles.